Vemuri Sivakumar

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Education

SRKR Engineering College — Bhimavaram, Andhra Pradesh
B.Tech in Artificial Intelligence and Machine Learning, CGPA: 8.32/10 — *2023 – 2027*

Sri Chaitanya Junior College — Rajahmundry, Andhra Pradesh
Higher Secondary (MPC), Percentage: 83% — *2021 – 2023*

Projects

AskCampus | *Python, LangGraph, RAG, FastAPI, Next.js, Qdrant, Groq, Gemini*

- Built a full-stack **RAG chatbot** that answers college queries from crawled website content using **LangGraph, Qdrant, FastAPI, and Next.js**.
- Improved retrieval quality using **hybrid search (vector + BM25)**, **Reciprocal Rank Fusion (RRF)**, and **cross-encoder reranking**, so exact names and course codes are found alongside semantically similar content.
- Designed an **agentic LangGraph workflow** that decomposes complex questions into sub-queries and rewrites queries when retrieved evidence is insufficient, improving retrieval for multi-part and ambiguous questions.
- Added **NeMo Guardrails** and evidence validation to mitigate prompt injection, verify faculty names and course codes against sources, and redact **PII** such as phone numbers.
- Implemented **SSE streaming**, response caching, **LLM API failover**, and **LangSmith tracing** to improve response latency, reliability, and observability.

Nebula Sketch | *Next.js, Node.js, WebSockets, Redis, Tailwind CSS, Docker*

- Built a real-time multiplayer drawing and guessing game where users join rooms, draw on a shared canvas, and compete by guessing words.
- Developed a server-side **Node.js game engine** managing round timers, word selection, hint generation, score calculation, and state synchronization.
- Streamed canvas stroke data over **WebSockets** for low-latency collaborative drawing, enabling real-time updates across all players in a room.
- Used **Redis** for room management, game state storage, and canvas persistence with auto-expiration, enabling auth-free room creation and joining.

REVA AI | *Python, Flask, React.js, PostGIS, AWS S3, REST APIs, JWT*

- Engineered a **voice-to-structured complaint pipeline** using Azure Speech and LLMs across multiple languages.
- Designed geospatial routing with **PostGIS** to automate jurisdiction mapping across **1,000+ locations**.
- Implemented secure authentication using **JWT, Aadhaar OTP (sandbox), PII masking, rate limiting**, and OWASP security practices.
- Built an evidence processing pipeline using **AWS S3** with duplicate image detection, transforming unstructured inputs into FIR-ready structured records and reducing processing time from hours to minutes.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL
AI / GenAI: RAG, LLMs, LangGraph, Prompt Engineering, NeMo Guardrails
Frameworks: FastAPI, Flask, React.js, Next.js, Tailwind CSS
Databases: PostgreSQL, Qdrant, MongoDB, MySQL, Redis, PostGIS
Tools: AWS S3, Docker, Git, GitHub, Linux, LangSmith
Core: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, REST APIs, WebSockets

Achievements

Smart India Hackathon 2024: 2nd place among 50+ teams as Team Leader
UDBHAV 2K26 (2026): Best Implementation Award for REVA AI at national-level technical fest
Coding Workshops (2024–2025): Organized 3 workshops for 100+ students on DSA and full-stack development
Python Certification – Cisco (2024): Completed Python Essentials certification by Cisco Networking Academy